Improving the PALM low-rank data splitter

Leakage minimization, controllable OOD difficulty, and n-D coverage

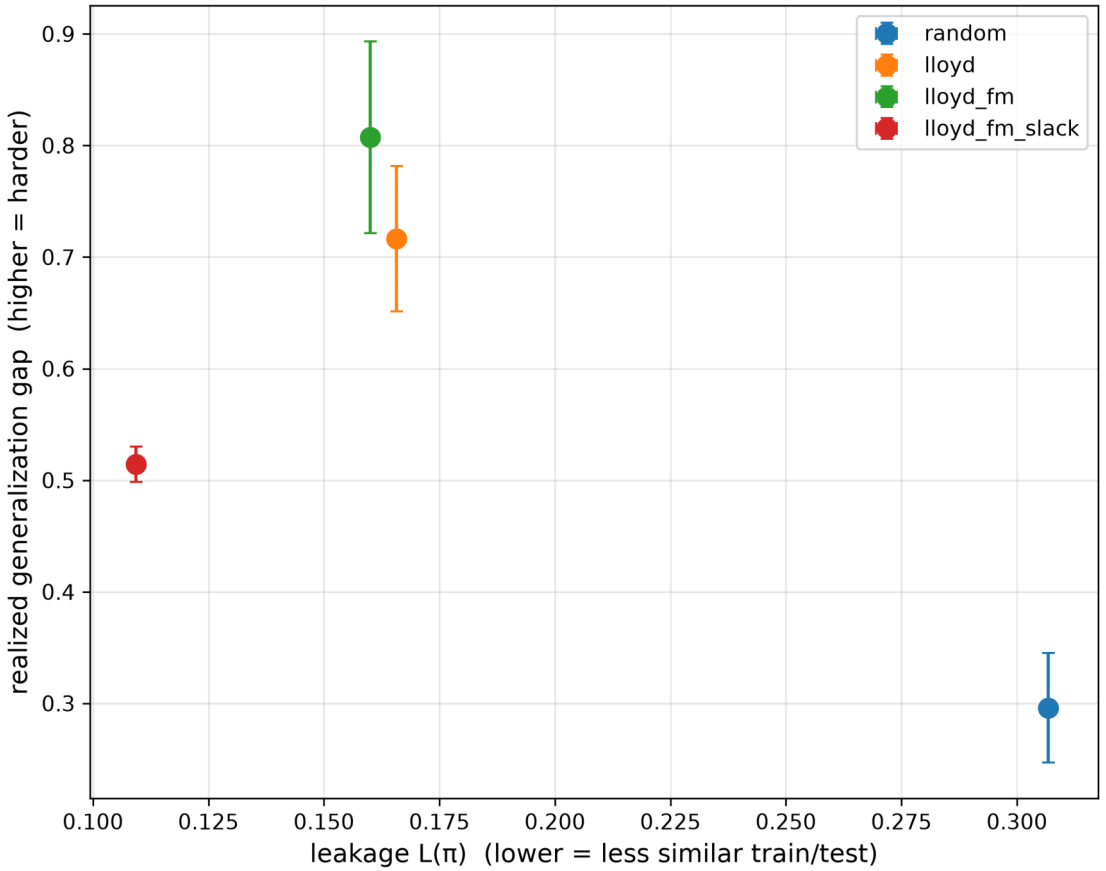
Ray Zhu

PALM · benchmark/master-suite

What was built

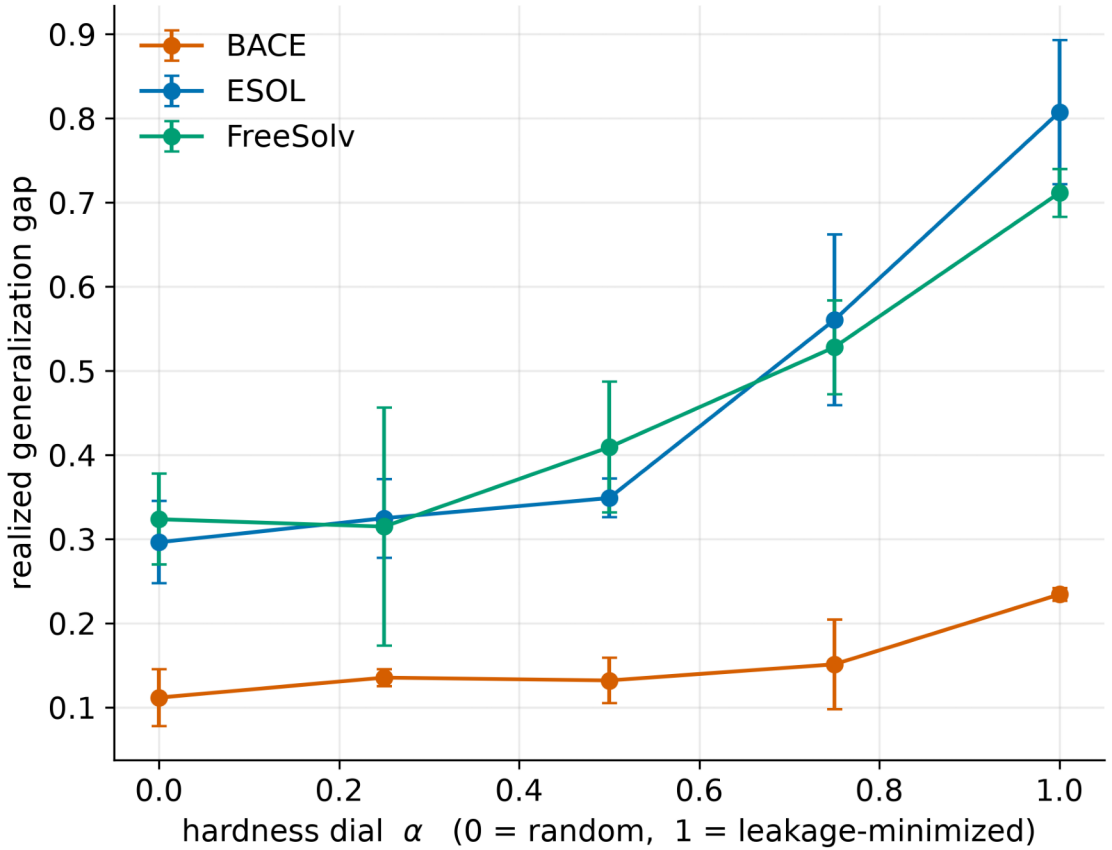
- **Scale FM cap fix**
  flat FM was already optimal; raising the 200k cap recovers ~4% leakage at scale
- **target_gap API**
  request a target difficulty; invert the hardness calibration to the split that delivers it
- **k>2 corridor**
  extend balance + hardness to 3-way train/val/test (−26–39% leakage)
- **Ablation study**
  consolidate every component on leakage AND generalization gap
- **lowrank_nd**
  n-D splitter — beats hypergraph on reactions; lowrank now covers 21/21 datasets

Leakage and difficulty pull apart



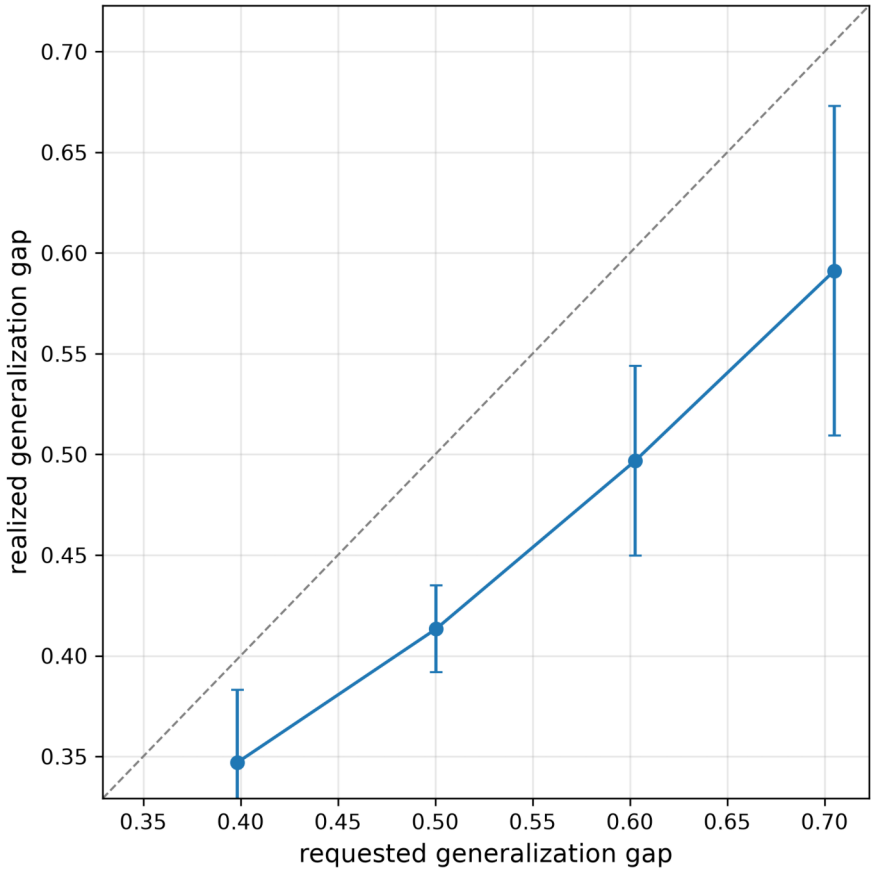
ESOL configs in leakage×gap space: random = easy corner; the exact-balance split is hardest; spending balance reaches the LOWEST leakage but also a LOWER gap — lower leakage the wrong way makes the eval easier.

A dial for OOD difficulty



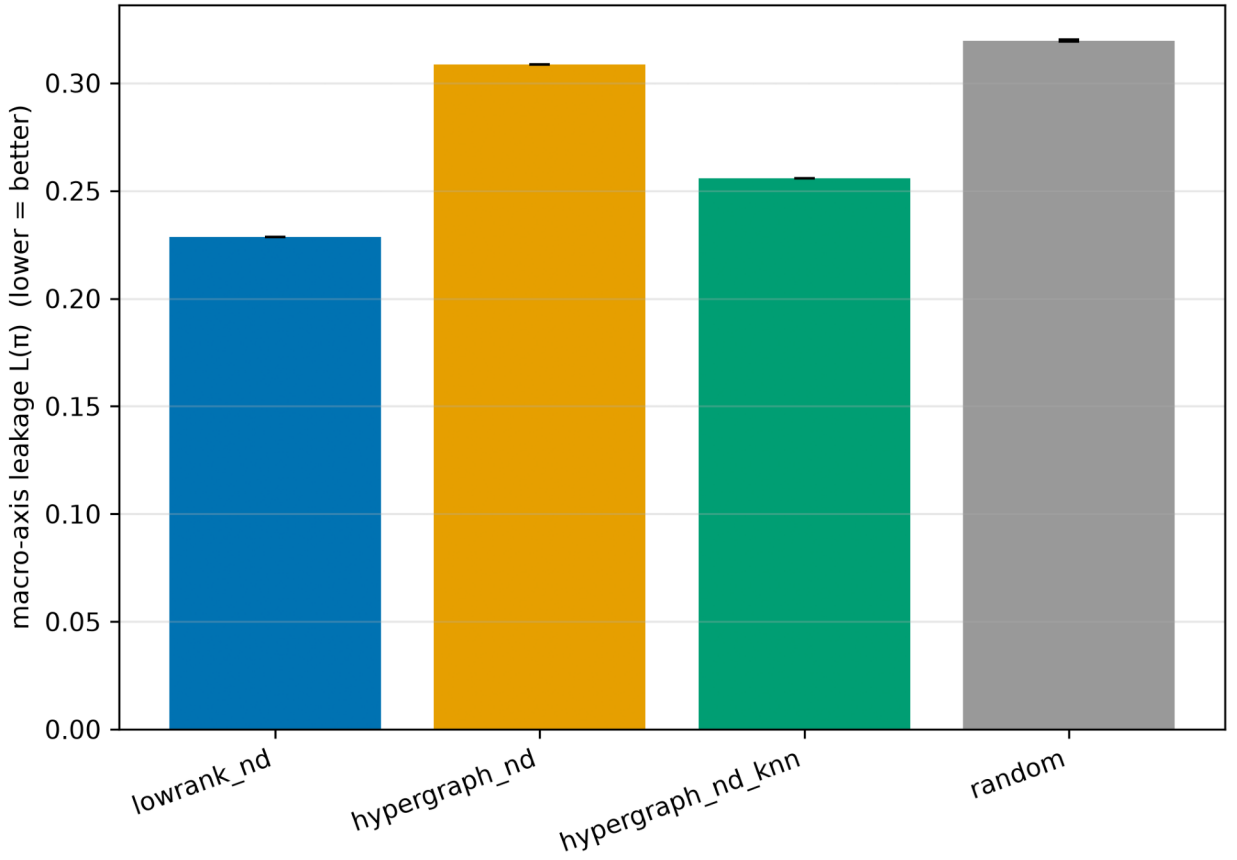
Realized train→test gap rises monotonically with the hardness dial α (Spearman +0.90 to +1.00 within a dataset).

Request a difficulty, get the split



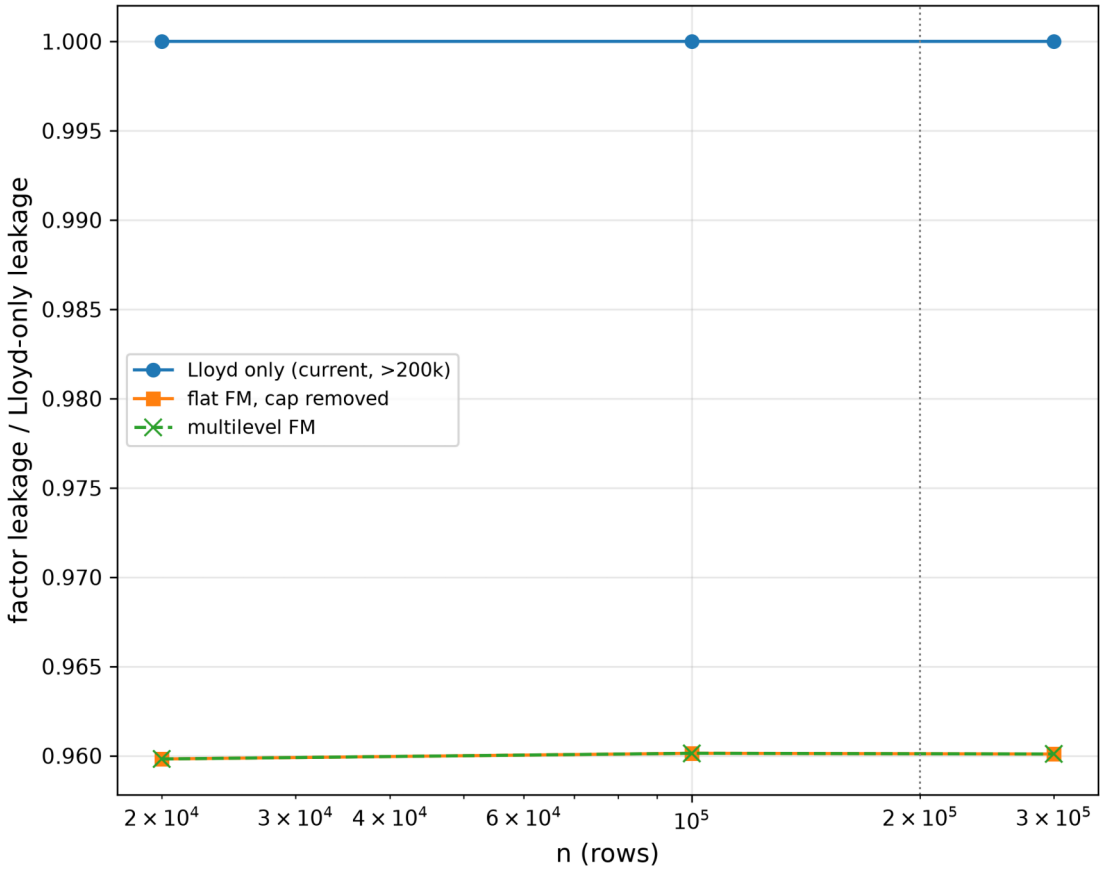
target_gap inverts the calibration: requested vs. realized gap on the y=x line (corr +0.90 to +0.997 across datasets).

lowrank_nd wins on reactions (n-D)



uspto_mcr macro leakage: lowrank_nd 0.229 (exact balance) vs hypergraph_nd_knn 0.256 (imbalanced) vs hypergraph_nd 0.309 (~96s) — lower, tighter, ~100× faster; lowrank now covers 21/21.

Recovering leakage at scale



Above n=200k the old FM cap shipped un-polished splits; raising it to 2M recovers ~4% leakage with flat FM (~12s at n=1M). The multilevel V-cycle matched it ~48× slower.

Key insights

- The generalization GAP — not the leakage number — is ground truth.
- Leakage & gap anti-correlate across mechanisms: the min-leakage split is NOT the hardest.
- The lever is the objective, not the solver — the optimizer is already at the global optimum.
- Difficulty is controllable (hardness) and now targetable (target_gap); the α→gap curve is convex.
- Rank / restarts are neutral on both metrics — only balance_slack & hardness are real levers.
- Directly optimizing the true objective wins at n-D: lowrank_nd beats hypergraph's proxy.